PROJECT DESIGN PHASE

Solution Architecture

Automated Network Request Management in ServiceNow

Date	Project Name	Maximum Marks
12 October 2026	Automated Network Request Management in ServiceNow	4 Marks

Solution Architecture

The solution uses standard ServiceNow catalog and workflow capabilities around a custom network request data model. Requesters interact with the Service Portal; Flow Designer coordinates record creation, approvals, notifications and updates; authorized network staff manage fulfilment using platform lists and records.

Logical architecture

Layer	Elements	Responsibility
Experience	Service Portal / Service Catalog	Request submission, user instructions, confirmation and status visibility.
Form logic	Catalog variables, variable sets, Catalog UI Policies	Capture structured inputs, validate required values and display conditional fields.
Orchestration	Flow Designer	Trigger on catalog submission; retrieve variables; create record; request approval; branch on outcome; send email; update state.
Data	Requested Item, custom Network Database table (e.g., u_network_database), approval records, fulfilment task table (e.g., u_network_task if implemented)	Maintain request details, source references, decisions, states and tasks.
Security	Table and field ACLs, roles, assignment/approval rules	Limit visibility and modification to authorized roles.
Operations	Flow execution details, system email logs, audit history, reports	Troubleshoot failures, confirm notifications and monitor lifecycle outcomes.

Lifecycle flow

Requester opens Network Request → completes catalog form → submits request → ServiceNow creates the Requested Item → Flow Designer retrieves catalog variables → creates the Network Database record → determines and requests approvals → approved path creates/updates fulfilment work and notifies participants → rejected path updates the request and notifies the requester → operators review status and audit history.

Architecture notes

- Keep the Requested Item reference on the custom record to support traceability.
- Configure the Approval Request related list only after confirming the relationship fields and table names in the instance; platform approval relationships can differ by configuration.
- Use least-privilege ACLs and test as requester, approver, fulfiller and administrator.
- Treat Cisco DNA Centre, Ansible and Integration Hub orchestration as future integrations unless configured and tested.